

NOTE Assume $\langle b \rangle$ aligned with the principal inertia axes then (*)

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$${}^b \underline{I}_c = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix}$$

Then the Euler Equation ($\underline{I}$ diagonal eq.)

$$\begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix} \begin{bmatrix} {}^b \dot{\omega}_1 \\ {}^b \dot{\omega}_2 \\ {}^b \dot{\omega}_3 \end{bmatrix} + \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix} \begin{bmatrix} {}^b \omega_1 \\ {}^b \omega_2 \\ {}^b \omega_3 \end{bmatrix} = \underline{M}_c^{6 \times T}$$

By computing the matrix products

$$\begin{cases} I_1 \dot{\omega}_1 + (I_3 - I_2) \omega_2 \omega_3 = ({}^b \underline{M}_c^{EXT})_1 \\ I_2 \dot{\omega}_2 + (I_1 - I_3) \omega_1 \omega_3 = ({}^b \underline{M}_c^{EXT})_2 \\ I_3 \dot{\omega}_3 + (I_2 - I_1) \omega_1 \omega_2 = ({}^b \underline{M}_c^{EXT})_3 \end{cases}$$

EXAMPLE (Dzhambekov effect)

Let $I_1 > I_2 > I_3 > 0$ (principal moments of inertia)
it is possible to notice that for $\underline{M}_c^{EXT} = 0$

if $\omega_i = \omega_j = 0$ then $\omega_k = \text{const.}$ is a solution of the Euler eqs.

It is possible to show that $\omega_1 = \text{const}$ and $\omega_3 = \text{const}$ correspond to "stable" rotations, while $\omega_2 = \text{const.}$ is unstable

(*) This condition is often exploited for bodies with symmetries (e.g. airplanes, drones, satellite-tes...)

CASE 1 Consider the equilibrium $\omega_1 = \text{const}$, $\omega_2 = \omega_3 = 0$ 22/
 Assume to slightly perturb ω_2 and ω_3 -
 The dynamics of ω_1 becomes

$$I_1 \dot{\omega}_1 = \underbrace{-(I_3 - I_2)}_{>0} \underbrace{\varepsilon_2 \varepsilon_3}_{\varepsilon} \rightarrow \dot{\omega}_1 \sim O(\varepsilon)$$

then $\omega_1 \approx \text{constant}$

Take the derivative of the second Euler eq.

$$I_2 \ddot{\omega}_2 = -(I_1 - I_3) \left[\omega_1 \dot{\omega}_3 + \dot{\omega}_1 \omega_3 \right]$$

$\uparrow$
 $-\left(\frac{I_2 - I_1}{I_3}\right) \omega_1 \omega_2$
negligible

Then $\ddot{\omega}_2 = + \frac{(I_1 - I_3)}{I_2} \cdot \frac{(I_2 - I_1)}{I_3} \omega_1^2 \omega_2 \approx -\gamma \omega_2$

stable dynamics >0 <0 $\gamma > 0$

NOTE $\ddot{\omega}_2 = -\gamma \omega_2 \Leftrightarrow \ddot{x} = -\gamma x \Rightarrow x(t) = \alpha \sin(\sqrt{\gamma} t)$

Some discussion for ω_3 -

CASE 3 Consider the equilibrium $\omega_3 = \text{const}$, $\omega_1 = \omega_2 = 0$ -
 [Discussion and conclusion as above]

CASE 2 Consider the equilibrium $\omega_2 = \text{const}$, $\omega_1 = \omega_3 = 0$
 Assume to perturb ω_1 and ω_3 -
 The dynamics of ω_2 becomes

$$I_2 \dot{\omega}_2 = -(I_1 - I_3) \varepsilon_1 \varepsilon_3 \Rightarrow \dot{\omega}_2 \sim O(\varepsilon) \Rightarrow \omega_2 \approx \text{const.}$$

Take the derivative of the first Euler eq.

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$$I_1 \ddot{\omega}_1 = - \underbrace{(I_3 - I_2)}_{\rightarrow} \left[\cancel{\dot{\omega}_2 \omega_3} + \omega_2 \dot{\omega}_3 \right] - \frac{I_2 - I_1}{I_3} \omega_1 \omega_2$$

negligible

Then

$$\ddot{\omega}_1 = \left(\frac{I_3 - I_2}{I_1} \right) \left(\frac{I_2 - I_1}{I_3} \right) \omega_2^2 \omega_1 = + \lambda \omega_1$$

$\lambda > 0$

unstable
dynamos
(*)

(*) The proof is not rigorous !!

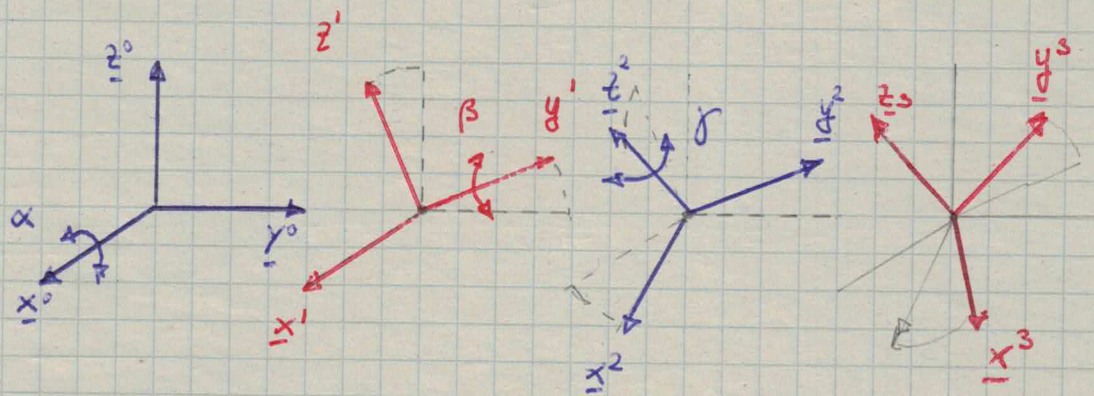
PARAMETERIZATION OF ANGULAR VELOCITY

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The orientation^(*) of a rigid body can be described by a sequence of rotations. These rotations can be defined as "elementary" rotations about frame axes.

There exist many possible choice of sequence of rotations described by the axis and corresponding angles. Euler angles (e.g. z-x-z) or rolls-pitch-yaw (x-y-z) are often used in practice.

ROLL-PITCH-YAW PARAMETERIZATION (**)



$${}^0R \triangleq {}^0R_x(\alpha) {}^1R_y(\beta) {}^2R_z(\gamma)$$

$R_e(\delta)$ ← rotation matrix about axis $\underline{e}$ by angle δ

(*) The notation is the same for other types of parameterization.

(*) a.k.a. attitude in aerospace jargon

$$R_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\alpha & -s\alpha \\ 0 & s\alpha & c\alpha \end{bmatrix}$$

NOTE $c\alpha \triangleq \cos \alpha$ 25/
 $s\alpha \triangleq \sin \alpha$

$$R_y(\beta) = \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix}$$

EXERCISE Compute 0R

$$R_z(\gamma) = \begin{bmatrix} c\gamma & -s\gamma & 0 \\ s\gamma & c\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$${}^0R = \begin{bmatrix} c\beta c\gamma & -c\beta s\gamma & s\beta \\ c\alpha s\gamma + s\alpha s\beta c\gamma & c\alpha c\gamma - s\alpha s\beta s\gamma & -s\alpha c\beta \\ s\alpha s\gamma - c\alpha s\beta c\gamma & s\alpha c\gamma + c\alpha s\beta s\gamma & c\alpha c\beta \end{bmatrix}$$

Notice that

$$\left. \begin{aligned} \underline{w}_{1/0} &= \underline{e}_1^0 \dot{\alpha} \\ \underline{w}_{2/1} &= \underline{e}_2^1 \dot{\beta} \\ \underline{w}_{3/2} &= \underline{e}_3^2 \dot{\gamma} \end{aligned} \right\} \Rightarrow \underline{w}_{b/0} = \underline{w}_{1/0} + \underline{w}_{2/1} + \underline{w}_{3/2}$$

↑
 additivity of
 angular velocities

Then

$$\underline{w}_{b/0} = \underline{e}_1^0 \dot{\alpha} + \underline{e}_2^1 \dot{\beta} + \underline{e}_3^2 \dot{\gamma} \quad (*)$$

$${}^0\underline{w}_{b/0} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \dot{\alpha} + \begin{bmatrix} 0 \\ c\alpha \\ s\alpha \end{bmatrix} \dot{\beta} + \begin{bmatrix} s\beta \\ -s\alpha c\beta \\ c\alpha c\beta \end{bmatrix} \dot{\gamma} \quad (**)$$

2nd column of
 ${}^0R = R_x R_y$

3rd column of
 0R

NOTE Computing α, β, γ from $\hat{\Omega}$

25.1/

$$\alpha = \text{atan} \left(-\frac{\kappa_{23}}{\kappa_{33}} \right)$$

$$\begin{array}{c} \uparrow \\ \frac{-(-s\alpha c\beta)}{c\alpha c\beta} = \frac{s\alpha}{c\alpha} \end{array}$$

$$\gamma = \text{atan} \left(-\frac{\kappa_{12}}{\kappa_{11}} \right)$$

$$\begin{array}{c} \uparrow \\ \frac{-(c\beta s\gamma)}{c\beta c\gamma} = \frac{s\gamma}{c\gamma} \end{array}$$

$$\beta = \text{atan} \left(\frac{\kappa_{13}}{\sqrt{1 - \kappa_{13}^2}} \right)$$

$$\frac{s\beta}{\sqrt{1 - s\beta^2}} = \frac{s\beta}{c\beta}$$

EXERCISE Compute ${}^b\omega_{b/o}$ [hint: use ${}^oR \dots$:) 26/

EXERCISE Compute $\frac{d}{dt} {}^o\omega_{b/o}$ in 2 ways:

- i) differentiate ~~(xx)~~ on page 25
- ii) apply ^{derivation and} Poisson formula in ~~(*)~~ " "

NEWTON-EULER EQUATIONS SUMMARY

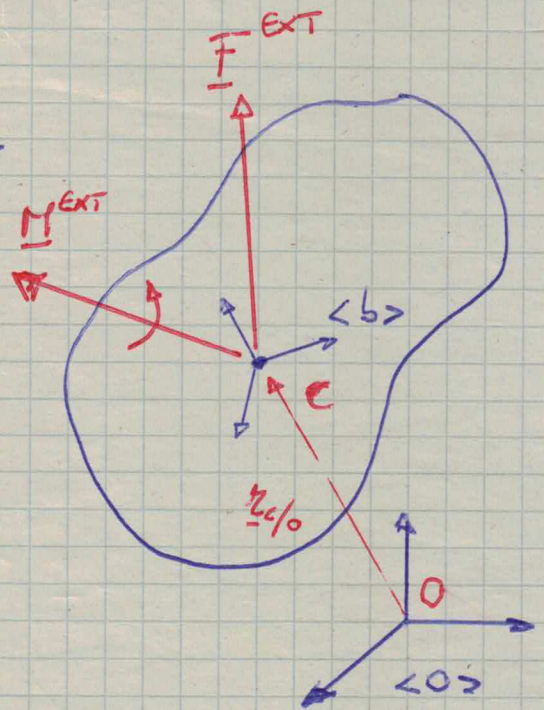
$$m \underline{a}_{c/o} = \underline{F}^{EXT}$$

$$\underline{I}_c(\underline{\dot{\omega}}_{b/o}) + \underline{\omega}_{b/o} \times \underline{I}_c(\underline{\omega}_{b/o}) = \underline{M}^{EXT}$$

coupled with

$$\frac{d}{dt} \underline{r}_{c/o} = \underline{v}_{c/o}$$

$$\frac{d}{dt} \underline{e}_i^b = \underline{\omega}_{b/o} \times \underline{e}_i^b$$



In algebraic (projected form)

$$m {}^o\underline{a}_{c/o} = {}^o\underline{F}^{EXT}$$

$${}^b\underline{I}_c {}^b\dot{\underline{\omega}}_{b/o} + [{}^b\underline{\omega}_{b/o} \times] {}^b\underline{I}_c {}^b\underline{\omega}_{b/o} = {}^b\underline{M}^{EXT}$$

coupled with

$$\frac{d {}^o\underline{r}_{c/o}}{dt} = {}^o\underline{v}_{c/o}$$

$${}^o\dot{\underline{R}} = {}^o\underline{R} [{}^b\omega_{b/o}]$$

More on momentum of forces

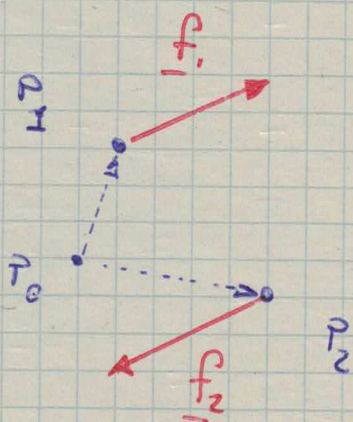
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Considers a couple of forces $\underline{f}_1$ and $\underline{f}_2$ applied at points P_1 and P_2 , s.t. $\underline{f}_1 + \underline{f}_2 = \underline{0}$ (the resultant total force is $\underline{0}$).

Consider the momentum of the forces w.r.t. a generic point P_0

$$\underline{m}_0 = (\underline{P}_1 - \underline{P}_0) \times \underline{f}_1 + (\underline{P}_2 - \underline{P}_0) \times \underline{f}_2 = \underline{z}_{1/0} \times \underline{f}_1 + \underline{z}_{2/0} \times \underline{f}_2$$

The condition $\underline{f}_1 + \underline{f}_2 = \underline{0} \Rightarrow \underline{f}_2 = -\underline{f}_1$



$$\underline{m}_0 = (\underline{z}_{1/0} - \underline{z}_{2/0}) \times \underline{f}_1 = \underline{z}_{1/0} \times \underline{f}_1$$

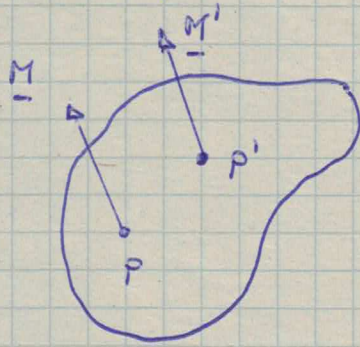
↑
invariant
w.r.t. P_0

a.k.a. torques

"Pure" moments $\underline{M}$ are generated by actuators like electric motors. The torques are invariant throughout the whole body.

Therefore $\underline{M}^{\text{ext}}$ in Euler eq. in general encompasses also torques as well as moments generated by external forces.

Pure moments (torques) are always invariant 28/
w.r.t. the point where they are expressed

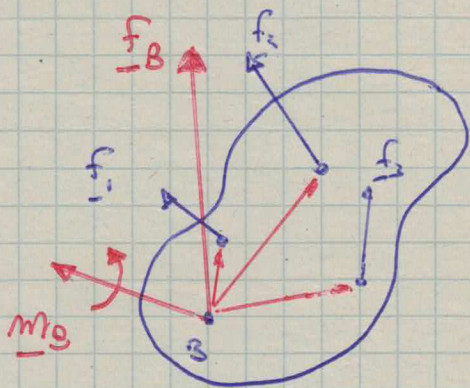


If $M = M'$ then the effect of M (or M') on the body motion is equivalent.

Given a set of forces $\underline{f}_1 \dots \underline{f}_L$ at points $P_1 \dots P_L$ their resultant effect on the body can be described as a single force and torque expressed w.r.t. a given point B.

$$\begin{cases} \underline{f}_B = \underline{f}_1 + \dots + \underline{f}_L = \sum_i \underline{f}_i \end{cases}$$

$$\begin{cases} \underline{m}_B = \underline{r}_{P_1/B} \times \underline{f}_1 + \dots + \underline{r}_{P_L/B} \times \underline{f}_L = \sum_i \underline{r}_{P_i/B} \times \underline{f}_i \end{cases}$$



Obviously $\underline{m}_B$ is a function of the point B ($\underline{f}_B$ is not) - (*)

(*) More on this in the next chapter.